

SOLOV'YEVSK, Russian Federation

WMO: 309670

Lat: 49.900N

Lon: 115.750E

Elev: 623

StdP: 94.06

Time Zone: 9.00 (E09)

Period: 86-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.7	-30.3	-35.0	0.1	-32.5	-32.7	0.2	-30.0	12.0	-18.4	9.6	-16.2	0.9	160	0.648

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	29.7	18.0	27.9	17.2	26.0	16.5	20.0	25.9	19.1	24.3	18.3	23.2	3.7	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	14.2	21.8	17.4	13.4	21.3	16.5	12.6	20.8	60.4	26.0	57.5	24.5	54.7	23.3	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.7	10.0	8.3	-36.7	33.1	3.3	1.5	-39.0	34.2	-40.9	35.0	-42.8	35.9	-45.2	37.0		
			-36.7	22.0	3.3	1.5	-39.1	23.0	-41.0	23.9	-42.8	24.7	-45.2	25.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	0.4	-22.4	-17.3	-7.9	2.5	10.7	16.8	19.7	18.1	10.5	2.0	-10.4	-18.6
	DBStd	15.23	4.28	4.57	5.99	4.46	4.80	3.89	2.60	2.64	3.95	4.58	5.74	4.50
	HDD10.0	4391	1006	763	554	227	49	3	0	0	41	249	612	886
	HDD18.3	6661	1264	997	812	475	241	72	15	36	237	506	862	1144
	CDD10.0	888	0	0	0	2	71	207	300	251	55	2	0	0
	CDD18.3	116	0	0	0	0	4	26	56	28	1	0	0	0
	CDH23.3	1137	0	0	0	0	113	296	445	266	15	2	0	0
	CDH26.7	281	0	0	0	0	27	72	107	73	2	0	0	0
Wind	WSAvg	3.0	2.0	2.3	3.1	4.3	4.0	3.7	3.0	2.6	3.2	3.1	2.9	2.2
Precipitation	PrecAvg	299	3	3	6	11	26	50	81	65	31	12	6	5
	PrecMax	577	9	9	22	34	54	100	216	170	75	34	20	14
	PrecMin	183	0	0	0	2	6	11	16	12	6	3	1	1
	PrecStd	90	2	2	5	7	13	24	43	39	17	7	5	3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-7.1	1.0	12.8	20.8	30.1	31.1	32.7	31.6	25.7	20.1	6.5	-4.8
		MCWB	-8.7	-2.2	4.6	9.8	15.8	17.3	19.9	19.8	14.9	9.9	1.3	-6.4
	2%	DB	-11.5	-4.3	7.8	16.5	26.7	28.7	29.6	28.8	22.4	15.0	3.0	-7.4
		MCWB	-12.5	-6.1	1.5	7.1	14.2	16.2	18.9	18.3	13.9	7.1	-1.0	-8.7
	5%	DB	-14.0	-7.3	4.3	13.8	23.5	26.5	27.7	26.2	19.8	12.1	0.1	-9.8
		MCWB	-14.7	-8.8	-0.7	5.4	12.4	15.9	17.7	17.5	12.8	5.7	-2.5	-10.9
	10%	DB	-15.8	-9.3	1.6	11.1	20.2	24.7	25.9	24.1	17.7	9.5	-2.2	-11.8
		MCWB	-16.4	-10.5	-2.1	3.8	10.5	15.0	17.5	16.5	11.7	4.0	-4.5	-12.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-8.6	-2.3	4.8	9.6	17.0	19.1	21.3	21.7	16.7	10.3	1.5	-6.3
		MCDB	-7.4	1.2	12.6	19.3	28.2	26.7	27.9	28.6	22.1	19.9	6.2	-4.9
	2%	WB	-12.5	-6.1	1.8	7.2	14.6	18.1	20.1	20.1	15.2	7.5	-0.8	-8.7
		MCDB	-11.6	-4.4	7.7	16.1	25.3	25.4	26.5	25.5	19.7	14.3	2.3	-7.4
	5%	WB	-14.7	-8.6	-0.5	5.5	12.7	17.2	19.5	18.9	13.9	5.9	-2.6	-10.9
		MCDB	-14.1	-7.3	3.9	13.1	22.3	24.3	25.0	23.7	18.2	11.1	0.0	-9.9
	10%	WB	-16.4	-10.4	-2.0	4.1	11.1	16.3	18.7	18.0	12.5	4.4	-4.4	-12.6
		MCDB	-15.8	-9.3	1.6	10.4	19.3	22.8	23.5	22.4	17.2	8.9	-2.3	-11.9
Mean Daily Temperature Range	5% DB	MDBR	10.3	12.9	12.7	13.1	13.8	11.7	10.1	10.1	10.3	11.3	10.4	9.9
		MCDBR	12.0	15.3	15.4	17.8	18.5	14.8	13.7	14.2	13.5	15.8	12.9	11.7
		MCWBR	11.2	13.3	10.2	10.0	8.7	5.9	5.1	5.9	6.6	9.3	9.4	10.5
	5% WB	MCDBR	12.0	15.2	14.8	16.7	16.8	12.7	10.5	11.2	10.9	15.1	12.7	11.4
		MCWBR	11.2	13.2	10.1	9.8	8.3	5.6	4.6	5.7	5.8	9.1	9.3	10.3
Clear-Sky Solar Irradiance	taub	0.254	0.280	0.320	0.404	0.484	0.463	0.423	0.389	0.340	0.310	0.278	0.262	
	taud	2.239	2.196	2.189	2.072	1.893	2.078	2.267	2.336	2.411	2.394	2.280	2.186	
	Ebn at Noon	795	858	880	834	778	801	831	845	856	819	752	713	
	Edh at Noon	71	96	117	149	186	156	128	113	94	79	68	65	
All-Sky Solar Radiation	RadAvg	1.70	2.89	4.38	5.42	6.16	6.43	5.94	5.31	4.24	2.89	1.72	1.27	
	RadStd	0.09	0.15	0.23	0.34	0.34	0.48	0.38	0.35	0.26	0.13	0.09	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	N/A	N/A	-2.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page